

Submission Summary

Conference Name

2026 International Conference on Intelligent and Sustainable Electronics & Computing Technologies

Track Name

Track 3: IoT, Artificial Intelligence & Machine Learning

Paper ID

1007

Paper Title

CipherDocs: A Secure Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification.

Abstract

Significant challenges exist in ensuring the security of electronic (digital) documents, including forgery, unauthorized alterations, inefficient verification mechanisms, and reduced public trust in conventional centralized document storage systems. This paper focuses on CipherDocs, its platform architecture, and the methodology adopted for the secure issuance and verification of documents. CipherDocs leverages blockchain technology (Polygon) and smart contracts to establish a robust data structure that preserves the integrity, authenticity, and transparent lifecycle management of critical records by permanently storing cryptographic hash values along with essential document metadata on the blockchain. To address scalability and storage limitations, the platform relies on off-chain, decentralized storage solutions for managing large-scale document data. Among other things, CipherDocs has dynamic QR codes that enable instant document verification; real-time authentication; and the automatic implementation of document expiration and revocation policies. CipherDocs offers a low-cost, secure, and flexible solution to document management issues by combining the immutability of blockchain technology with simplified verification methods and automated lifecycle management, thus resulting in a much higher degree of trust and prevention of trust in the digital government systems of the future.

Created

1/30/2026, 11:24:43 PM

Last Modified

1/30/2026, 11:24:43 PM

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Primary Subject Area

Data-Driven Optimization & Autonomous Systems

Secondary Subject Areas

Other work related to IOT, Artificial Intelligence & Machine Learning

Submission Files

CipherDocs A Secure Solution to Prevent Forgery of Critical and Official Documents Using Polygon Blockchain-Based Encryption and Verification.pdf (345.9 Kb, 1/30/2026, 11:22:58 PM)
